Terms

Read these first. Three are used more narrowly here than in ordinary usage, and every confusing number below traces back to one of them.

- **A lesion is** an area of abnormal tissue, so for us a tumor: one three-dimensional mass a radiologist would put in a report. **In our metrics a lesion is** something much narrower — one connected blob of tumor mask within a *single 2D slice*. A scan is a stack of about 155 cross-sections, so one tumor crossing twenty of them counts as twenty lesions, a thin or broken enhancing rim fragments into several more inside each slice, and a five-pixel speck counts the same as a whole tumor.
- **Erasure is** the failure we set out to fix. **The erasure rate is** the number of true lesions the model failed to recover, divided by the number of true lesions, where a lesion counts as recovered if the prediction covers at least 10% of it. Lower is safer.
- **Hallucination is** the opposite failure: the reconstruction invents tumor where there is none. **The hallucination rate is** the number of predicted blobs that are fabricated, divided by the number of predicted blobs, where a blob counts as fabricated if less than 10% of it sits inside real tumor. Lower is safer, and it tends to move against erasure.
- **Dice is** a measure of how much two outlines overlap, from 0 for none to 1 for identical. We report it because it is how we check the two models are compared at equal segmentation quality: if ours simply segmented better across the board, the difference in erasure could not be credited to the training objective.
- **The floor is** what the tumor detector misses when handed the untouched original scan, with no degradation or reconstruction involved. It is large, so it is the denominator every erasure rate must be read against.
- **The baseline is** the distortion-optimal model: the standard approach, trained only to minimise average pixel error. **Ours is** the tumor-aware model: the same network and the same data, with errors inside the tumor weighted far more heavily.

The result

Super-resolution on a cheap blurry scan can quietly delete a small tumor, because the model is graded on average pixel error and a small tumor is a tiny fraction of the pixels. We changed what the model is punished for, weighting errors inside the tumor far more heavily, and then measured how much tumor survives. On 70 held-out BraTS patients it loses far less tumor than the standard model, at the *same* segmentation quality.

What the tumor detector was given	Tumor missed
The low-resolution scan, as a cheap scanner produces it	62.2%
After standard super-resolution (baseline)	58.0%
After tumor-aware super-resolution (ours)	**51.3%**

Super-resolution helps, and ours helps most. 66 of 70 patients improve, 1 is worse, $p < 0.0001$, and quality is genuinely matched rather than approximately: Dice 0.660 against 0.675, and brain-masked PSNR within 0.23 dB, so ours is not simply producing sharper pictures.

Standard super-resolution recovers some of what the low-resolution scan loses, and ours recovers substantially more: **6.7 points fewer lesions missed than the baseline**, at matched quality.

One caveat to have ready, because it is the first thing a careful reader will ask. These rates are not “half the tumors disappear”: the tumor detector we measure through is itself imperfect, and on the untouched original scan it already misses 47.6% of these small components. Most of every figure in the table is that detector rather than the super-resolution, which is also why the improvement from 58.0% to 51.3% is larger than it looks — it is 6.7 points out of the roughly 10 that super-resolution is responsible for at all.

Is a rate near 50% bad? Split it by lesion size

Lesion size	How many	Low-resolution	Baseline	Ours
Small, < 50 px	6,762 (71%)	82.1%	78.6%	**69.8%**
Medium, 50–200 px	1,005	30.0%	16.4%	**13.5%**
Large, > 200 px	1,723	3.2%	1.2%	**0.9%**

Start with the bottom row: a large lesion is missed under 1% of the time by either model, so substantial tumors are not vanishing. The overall rate is high because 71% of all components are under 50 px, and the detector misses 65.1% of those even on the untouched original scan. Small does not mean unimportant — a 30 px enhancing focus can be an early recurrence, and this is where the remaining work is — but it is also partly an artefact of counting a fragmenting rim as many equal lesions.

Three things that make it credible, and one that limits it

- **It appears only where the theory says it must.** The benefit is there for the small enhancing tumor and *vanishes* for the whole tumor region, nine times larger. That is the prediction: the objective is only misaligned when the structure is too small to matter to a pixel-error score.
- **The patient is the unit of analysis.** One tumor spans many slices and appears as a lesion in each, so those outcomes are correlated; pooling them fakes an interval about 1.5 times too tight. The table above pools every cross-section, which is fine for reading off rates but not for a confidence interval. Computing one rate per patient and resampling patients gives a mean paired difference of 6.5 points, 95% CI $[5.2, 8.2]$.
- **The weakest link is the detector, not the super-resolution.** It misses 45.5% of lesions on an untouched scan. A stronger downstream segmenter would buy more than a better loss function.
- **The limit: it is not free.** Protecting tumors makes the model readier to report one that is not there, and hallucination rises from 0.266 to 0.387. A second loss term recovers most of that cost for about half the benefit, so the tradeoff is a dial rather than a fixed price.

Provenance: enhancing-tumor checkpoint trained on 17,233 axial slices from 468 BraTS patients (Medical Segmentation Decathlon brain task), split by patient, measured on the 70-patient validation split. Raw outputs in results/*; checkpoints at* douyeszn/tumor-aware-sr *on Hugging Face. The proposal's results table is from an earlier, smaller run, so its digits differ.*